Group 29- Literature Review table

Table 1- Articles group numbers

Group No	Articles
G1	**AI-Powered Self-Healing Cloud Infrastructures: A Paradigm For Autonomous Fault Recovery** Ravi Kumar Vankayalapati1 , Chandrashekar Pandugula2 , Venkata Krishna Azith Teja Ganti3 , Ghatoth Mishra4
G2	**An Intelligent Fault Self-Healing Mechanism for Cloud AI Systems via Integration of Large Language Models and Deep Reinforcement Learning** 1 st Ze Yang , 2 nd Yihong Jin, 3 rd Juntian Liu, 4th Xinhe Xu
G3	**Self-Healing Networks with AI-Based Fault Prediction in IoT Ecosystems** Manohar Jain
G4	**AI-powered self-healing enterprise applications: A new era of autonomous systems** Praveen Kumar Guguloth
G5	**AI-Driven Infrastructure Automation: Leveraging AI and ML for Self-Healing and Auto-Scaling Cloud Environments** Ali Asghar Mehdi Syed, Erik Anazagasty
G6	**Building AI Agents for Autonomous Clouds: Challenges and Design Principles** Manish Shetty2, Yinfang Chen3, Gagan Somashekar1, Minghua Ma1, Yogesh Simmhan4, Xuchao Zhang1, Jonathan Mace5, Dax Vandevoorde6, Pedro Las-Casas1, Shachee Mishra Gupta1, Suman Nath5, Chetan Bansal1, Saravan Rajmohan1
G7	**Artificial intelligence–enabled self-healing infrastructure systems** Lauren McMillan
G8	**Smart Security for Smart Clouds: Real-Time AI Defense and Self Healing Architecture Integration** Asia Rehman, Grayson Adrian

G9	**Self-Healing AI Systems: Autonomous Detection and Recovery from Security Breaches** Holmes Walter
G10	**Intelligent Cloud Resilience: Self-Healing Architectures and AI Driven Threat Automation** Samiullah Khan, Nadeem Abbas

Propose Study - **Autonomous AI Bot for Self-Healing and Proactive Server Health Management in Distributed Systems**

Table 2- Factor Summary table

Factors	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10	Propose Study
Autonomous Detection	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Self-Healing Mechanism	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Adaptive Learning	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Explainable Healing Actions	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
Adversarial-Aware Security	✗	✗	✓	✗	✓	✗	✗	✓	✓	✓	✓
Federated Self-Healing	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓
Human-in-the-Loop Option	✓	✗	✓	✓	✓	✓	✓	✓ Partial	✗	✓ Partial	✓
Multi-Layered Security Monitoring	✗	✗	✓	✗	✓	✓	✗	✓	✓	✓	✓
Simulation Testbed for Cyber-Attacks and Healing	✗	✓	✓	✗	✗	✓	✓	✗	✓	✗	✓
Performance Metrics and Reporting Dashboard	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Key Features of Our Research Project

1. **Autonomous Threat Detection**
   - Real-time detection of cyber threats (e.g., intrusions, malware, anomalies) using machine learning (ML) models.
   - Behavior-based analysis for accuracy beyond signature-based methods.
2. **Self-Healing Mechanism** ○ Automatic containment (e.g., isolating affected system parts). ○ Recovery through rollback, patching, or restarting affected services without human intervention.
3. **Adaptive Learning**
   - The system continuously improves itself by learning from past attacks and healing events (online learning).

4. **Explainable Healing Actions (NEW)**
    - Uses explainable AI (XAI) to provide human-readable justifications for detection and healing decisions.
    - Increases trust, helps administrators understand automated decisions.

5. **Adversarial-Aware Security (NEW)**
    - Protects against attackers who try to deceive AI models (e.g., adversarial ML attacks).
    - Ensures healing actions are not exploited or triggered wrongly.
6. **Federated Self-Healing (NEW)**
    - Enables multiple systems/devices (IoT, edge) to collaborate in detecting and healing threats while preserving privacy.
    - Uses federated learning for decentralized intelligence sharing.
7. **Human-in-the-Loop Option (NEW)**
    - Allows administrators to approve or override critical healing actions.
    - Balances automation with control in sensitive environments.
8. **Multi-Layered Security Monitoring (NEW)**
    - Monitors at multiple levels: system logs, network traffic, and application behavior.
    - Increases detection coverage and healing precision.
9. **Simulation Testbed for Cyber-Attacks and Healing (NEW)** ○ Develops a controlled environment to simulate attacks and test healing responses.
    - Useful for training, evaluation, and future research.
10. **Performance Metrics and Reporting Dashboard (NEW)**
    - Real-time reporting of system health, threat statistics, and healing effectiveness (e.g., Mean Time To Heal, false positives).
    - Visual dashboard for monitoring system status and actions taken.